

TPC3

Research Questions

- Base on my research topic: **The effects of computer games on players, particularly in three main aspects: addiction, aggression and cognitive influence.**
- I refined my topic into several research questions below:
 1. What are the psychological and behavioural impacts of video game addiction on adolescents and young adults?
 2. How does exposure to violent video games influence aggression and social behavior in players?
 3. What are the cognitive benefits and drawbacks of regular video gaming in terms of problem-solving, multitasking, and attention span?
 4. How do different game genres affect mental health outcomes in players?
 5. What are the most effective intervention strategies to mitigate the negative effects of video game addiction and aggression while preserving their cognitive and social benefits?

Systematic Research

- For my systematic research, I am still going to use **Google Scholar, UTSC Library, PsycINFO, ACM Digital Library, IEEE Xplore, and PubMed** to retrieve relevant academic papers.
- Below are the relevant queries and applied filter:

Search Queries

1. "video game addiction" AND "compulsive gaming" AND "mental health" after:2015
2. "gaming disorder" AND "WHO classification" AND "behavioural effects"
3. "violent video games" AND "aggression" AND "meta-analysis" after:2015
4. "video games" AND "cognitive flexibility" AND "reaction time improvement"
5. "gaming" AND "attention span" AND "cognitive benefits OR cognitive impairment"
6.

Applied Filter

1. Year: 2015–Present
2. Peer-reviewed journal articles only

3. Studies with sample sizes greater than 100 participants

Database

1. Google Scholar
2. UTSC Library
3. PsycINFO
4. ACM Digital Library
5. IEEE Xplore

Selected Papers & Bibliography

- I will put all reference for each source at the end of this paper, so there I only give the author and title.
1. Shams et al. (2015) – Effects of video games on cognition and brain structure.
 2. Zhang et al. (2021) – Aggressive cognition and violent video games.
 3. Li et al. (2018) – Mindfulness-oriented recovery enhancement for video game addiction.
 4. Kneer et al. (2016) – Relationship between game difficulty, violence, and aggression.
 5. Sauer et al. (2015) – Narrative context and reward structures in violent games.
 6. Bediou et al. (2018) – Meta-analysis of Action Video Game Impact on Perceptual, Attentional, and Cognitive Skills
 7. Prescott et al. (2018) – Meta-analysis of the Relationship Between Violent Video Game Play and Physical Aggression Over Time
 8. Johannes et al. (2022) – Video Game Violence, Aggression, and Delinquency: A Registered Report
 9. Przybylski & Weinstein (2019) – Violent Video Game Engagement is Not Associated with Adolescents' Aggressive Behaviour
 10. Li et al. (2021) – The Effects of Gaming Disorder on Adolescent Sleep: A Longitudinal Study

Unsystematic Research

- Since all systematic search are primarily academic papers, I extend my search range to others, such as the news article, blog posts, government reports and so on.
- I will put all reference for each source at the end of this paper, so there I only give the author and title.
- Below are some of my search result :
 1. The Australian (2023) – “Gaming Linked to Problem Gambling”
 2. The Times (2023) – “‘Phone Dope’ Is the Addiction of Our Times”

3. The Guardian (2024) – “Totally Wired: Why Are So Many Young People Addicted to Video Games?”
4. FOX 26 Houston (2023) – “Are Video Games Making Kids Violent? FOX 26 Investigates”
5. New York Post (2023) – “‘Video-Game Addiction Ruined My Life’”
6. American Psychological Association (2000) – “Violent Video Games Can Increase Aggression”
7. NPR (2023) - “After Riots in France, Macron Partially Blames Video Games for Violence”
8. ABC News - (2023“Mom Claims Video Game Companies Are Intentionally Addicting Kids”)
9. University of Oxford (2019) - “Violent Video Games Found Not to Be Associated with Adolescent Aggression”
10. Psychiatric Times (2020) - “Is Video Game Addiction a Disorder?”

CRAAP Analysis

- For each selected source, I use CRAAP analysis to illustrate why they are relevant to my topic.
- I will put all reference for each source at the end of this paper, so there I only give the author and title.
- Here are 5 systematic sources and 5 unsystematic sources:
 1. **Shams et al. (2015) – Effects of Video Games on Cognition**
 - **Currency:** Published in 2015, the study remains relevant for foundational insights into gaming’s cognitive effects.
 - **Relevance:** Directly examines how video games impact cognition, aligning with research question 4.
 - **Authority:** Authored by T.A. Shams et al., and published in the peer-reviewed *Current Psychiatry Reports*.
 - **Accuracy:** Reviews studies from 2011 to 2014, indicating that specific video games can alter brain structure and enhance cognitive functions.
 - **Purpose:** Aims to inform the scientific community about potential cognitive and structural brain changes associated with video gaming.
 2. **Zhang et al. (2021) – Aggressive Cognition & Violent Games**
 - **Currency:** Published in 2021, this study offers recent insights into the effects of violent video games on aggression.
 - **Relevance:** Directly examines the impact of violent video games on aggressive cognition and behavior in children, aligning with research question 1.

- **Authority:** Authored by Qian Zhang, Yi Cao, and JingJin Tian, and published in the peer-reviewed journal.
 - **Accuracy:** Utilizes a sample of 300 children, employing experimental methods to demonstrate that brief exposure to violent video games increases aggressive cognition and behavior, with a more pronounced effect in boys.
 - **Purpose:** Inform the scientific community about the relationship between violent video game exposure and aggression in children, providing evidence to support interventions targeting aggression, especially in boys.
3. **Li et al. (2018) – Mindfulness-Oriented Recovery Enhancement for Video Game Addiction**
- **Currency:** Published in 2018, this study aligns with contemporary discussions on gaming disorders.
 - **Relevance:** The research focuses on treating video game addiction, aligning with research question 5.
 - **Authority:** Authored by mental health professionals and published in the *International Journal of Mental Health and Addiction*.
 - **Accuracy:** Based on empirical case studies, the study provides preliminary evidence supporting the effectiveness of Mindfulness-Oriented Recovery Enhancement (MORE) in reducing video game addiction and enhancing positive coping mechanisms.
 - **Purpose:** Inform and guide clinicians and researchers by presenting an adapted mindfulness treatment protocol for video game addiction, emphasizing the need for further research and offering practical insights for intervention development.
4. **Kneer et al. (2016) – Game Difficulty & Aggression**
- **Currency:** Published in 2016, this study remains relevant as it examines the psychological effects of game mechanics, a topic still widely researched.
 - **Relevance:** It explores how game difficulty, rather than violent content, aligning with question 1.
 - **Authority:** Authored by experienced researchers in game psychology and published in *Computers in Human Behavior*.
 - **Accuracy:** The study employs controlled experiments with validated psychological measures to assess aggression levels, ensuring reliability.
 - **Purpose:** The research aims to contribute to academic discussions on game mechanics and aggression, maintaining a neutral and analytical stance.
5. **Sauer et al. (2015) – Narrative & Reward Structures in Violent Games**
- **Currency:** Published in 2015, this study provides foundational insights into how narrative and reward structures in violent games impact player aggression.
 - **Relevance:** It directly examines the relationship between in-game reward mechanics and aggression, aligning with research question 1 & 2.

- **Authority:** Conducted by experts in psychology and gaming studies, the research is published in a well-regarded, peer-reviewed psychology journal.
- **Accuracy:** The study relies on behavioral experiments and empirical data to assess how different game structures influence aggression levels.
- **Purpose:** The research aims to objectively analyze how game design elements shape player behavior, without advocating for or against violent games.

6. **The Australian (2023) – “Gaming Linked to Problem Gambling”**

- **Currency:** Published in 2023, this article presents recent findings on the association between video game loot boxes and gambling behaviors.
- **Relevance:** The article discusses the potential mental health implications of in-game purchases, relating to question 1.
- **Authority:** Authored by journalists from *The Australian*, a reputable news outlet, and references an international study published in *Computers in Human Behavior*.
- **Accuracy:** The article cites empirical research involving 1,400 adult gamers, providing data on the link between loot boxes and problem gambling.
- **Purpose:** Aims to inform the public about the potential risks associated with certain gaming practices, particularly loot boxes.

7. **The Times (2023) – “‘Phone Dope’ Is the Addiction of Our Times”**

- **Currency:** Published in 2023, the article addresses the contemporary issue of smartphone and gaming addiction.
- **Relevance:** Explores the addictive nature of digital activities, including gaming, and their impact on mental health.
- **Authority:** Written by journalists from *The Times*, a well-established news organization.
- **Accuracy:** The article references legislative efforts and expert opinions on digital addiction but lacks empirical data.
- **Purpose:** Seeks to raise awareness about digital addiction and advocate for policy changes to address the issue.

8. **The Guardian (2024) – “Totally Wired: Why Are So Many Young People Addicted to Video Games?”**

- **Currency:** Published in 2024, this article examines the increasing prevalence of video game addiction among youth.
- **Relevance:** Directly addresses concerns about gaming addiction in young populations, aligning with question 4.
- **Authority:** Authored by journalists from *The Guardian*, a respected international news outlet.
- **Accuracy:** Includes interviews and anecdotal evidence but lacks comprehensive statistical data.

- **Purpose:** Aims to shed light on the factors contributing to gaming addiction among young people and its societal implications.
9. **FOX 26 Houston (2023) – “Are Video Games Making Kids Violent? FOX 26 Investigates”**
 - **Currency:** Published in 2023, the article investigates the potential link between violent video games and real-world aggression.
 - **Relevance:** Addresses the debate on whether violent video games contribute to aggressive behavior in children.
 - **Authority:** Produced by FOX 26 Houston, a regional news station.
 - **Accuracy:** References expert opinions and existing research, suggesting minimal evidence supporting the claim that violent video games cause real-world violence.
 - **Purpose:** Aims to inform viewers about the current understanding of the relationship between violent video games and aggression.
 10. **New York Post (2023) – “Video-Game Addiction Ruined My Life”**
 - **Currency:** Published in 2023, the article presents personal accounts of individuals affected by video game addiction.
 - **Relevance:** Highlights the real-world consequences of excessive gaming, contributing to the discourse on gaming addiction.
 - **Authority:** Authored by journalists from the *New York Post*, a widely read news outlet.
 - **Accuracy:** Based on personal testimonies, providing anecdotal evidence without empirical data.
 - **Purpose:** Aims to raise awareness about the potential dangers of uncontrolled gaming habits.

Background

- Below are **summaries of key sources** from my systematic and unsystematic research, structured to address the three main themes: addiction, aggression, and cognitive influence.

Video Game Addiction

1. **Shams et al. (2015) – Effects of Video Games on Cognition and Brain Structure**
 - **What it does:** Investigates the neurological effects of video games on brain function.
 - **Findings:** Certain video games can alter brain structure and cognitive functions.
 - **Relevance:** Provides a foundation for understanding addiction mechanisms and potential cognitive enhancements.
2. **Li et al. (2018) – Mindfulness-Oriented Recovery Enhancement for Video Game Addiction**

- **What it does:** Proposes mindfulness-based recovery techniques for video game addiction.
 - **Findings:** Shows preliminary success in reducing addiction through mindfulness interventions.
 - **Relevance:** Directly supports research question 5 (interventions for addiction mitigation).
3. **The Guardian (2024) – “Totally Wired: Why Are So Many Young People Addicted to Video Games?”**
- **What it does:** Discusses the increasing prevalence of video game addiction.
 - **Findings:** Identifies social and psychological factors contributing to addiction.
 - **Relevance:** Helps contextualize gaming addiction as a societal issue beyond academic research.

Video Games and Aggression

1. **Zhang et al. (2021) – Aggressive Cognition & Violent Games**
- **What it does:** Examines how violent video games impact aggression in children.
 - **Findings:** Increases in aggressive cognition and behavior, especially in boys.
 - **Relevance:** Aligns with research question 2 on aggression.
2. **Kneer et al. (2016) – Game Difficulty & Aggression**
- **What it does:** Investigates whether game difficulty or violent content affects aggression.
 - **Findings:** Game difficulty, rather than violence, is more likely to increase aggression.
 - **Relevance:** Challenges traditional views on the link between violent games and aggression.
3. **Przybylski & Weinstein (2019) – Violent Video Games and Adolescent Aggression**
- **What it does:** Analyzes whether violent video game engagement leads to aggression.
 - **Findings:** No significant correlation between video games and increased aggression.
 - **Relevance:** Provides a counterpoint to studies suggesting a strong link between gaming and aggression.
4. **FOX 26 Houston (2023) – “Are Video Games Making Kids Violent?”**
- **What it does:** Investigates real-world effects of video games on violence.
 - **Findings:** Minimal evidence supporting claims that video games directly cause violence.
 - **Relevance:** Provides a media perspective on the aggression debate.

Cognitive Effects of Video Games

1. **Bediou et al. (2018) – Meta-Analysis of Action Video Games’ Impact on Cognitive Skills**

- **What it does:** Reviews the impact of video games on perception, attention, and cognition.
- **Findings:** Action video games improve cognitive flexibility, reaction time, and multitasking.
- **Relevance:** Supports research question 3 on cognitive benefits.

2. Sauer et al. (2015) – Narrative & Reward Structures in Violent Games

- **What it does:** Studies how in-game rewards and narratives influence player behavior.
- **Findings:** Different game structures can shape player cognition and decision-making.
- **Relevance:** Relates to **how game mechanics influence cognitive development**.

3. University of Oxford (2019) – Violent Video Games and Adolescent Aggression

- **What it does:** Examines long-term effects of violent games.
- **Findings:** No consistent evidence that violent video games impact adolescent behavior negatively.
- **Relevance:** Contrasts with other studies on aggression and cognitive impact.

Paper Outline

Introduction

- Define the problem: The impact of video games on addiction, aggression, and cognition.
- Present research questions.
- Explain why this topic is important (psychological, social, and technological implications).

Background

- Summarize previous research (as outlined above).
- Highlight key debates (e.g., do violent games cause aggression?).

The Impact of Video Game Addiction

- Discuss psychological effects (dependency, withdrawal symptoms).
- Explain behavioral consequences (academic performance, social isolation).
- Explore intervention strategies (mindfulness, parental control, policy recommendations).

Video Games and Aggression

- Present studies supporting the aggression hypothesis (Zhang et al., 2021).
- Present contradictory findings (Oxford University, 2019).
- Discuss factors beyond violence (e.g., game difficulty affecting frustration).

Cognitive Benefits and Drawbacks

- How video games improve cognitive skills (problem-solving, attention span, multitasking).
- Potential negative effects (shortened attention spans, reduced impulse control).
- Compare findings from neuroscientific and psychological studies.

Mental Health and Video Game Genres

- How different genres (action, RPG, strategy) impact stress, anxiety, and depression.
- Review psychological theories explaining gaming as escapism vs. cognitive enhancement.

Interventions and Policy Recommendations

- Review regulatory approaches (age restrictions, warning labels).
- Discuss mental health interventions (therapy, school-based programs).
- Evaluate gaming industry responsibility in promoting healthier gaming habits.

Conclusion

- Summarize key findings.
- Answer research questions.
- Provide personal insights and future research suggestions.

Reference

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